LUIS HUMBERTO MOTA REYES

Mechanical Engineering Student

SolidWorks | Fusion 360 | FEA | CFD | Manufacturing

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PROFILE

Mechanical Engineering student at Tecnológico de Monterrey with experience in mechanical design, CAD modeling, prototype development, manufacturing, and engineering analysis for aerospace and robotics projects. Currently leading the Design & Manufacturing Team at Ignitia Rocket Team, overseeing the design, simulation, manufacturing, and validation of experimental rocket components using SolidWorks, Fusion 360, FEA, and CFD.

Passionate about product development, manufacturing processes, and mechanical design, with strong analytical skills, initiative, and the ability to quickly learn new engineering tools while collaborating effectively in multidisciplinary teams.

ENGINEERING EXPERIENCE

Ignitia Rocket Team | *Design & Manufacturing Team Leader*

- Lead the Design & Manufacturing team responsible for the development of experimental rockets.
- Design mechanical components and assemblies using SolidWorks and Fusion 360.
- Perform Finite Element Analysis (FEA) and Computational Fluid Dynamics (CFD) simulations.
- Design aluminum and composite components considering manufacturability and structural performance.
- Coordinate manufacturing and assembly processes; collaborate with propulsion, avionics, and recovery teams.
- Participate in prototype validation through static fire testing and flight preparation.

FIRST Robotics Competition — FRC Daedalus 7102 | *Team Captain*

- Led multidisciplinary engineering projects from concept to competition.
- Coordinated mechanical design and system integration.
- National Champion 2025; qualified for the FIRST World Championship.

FIRST Tech Challenge — FTC Ikarus | *Founder & Team Captain*

- Founded and managed the robotics team; coordinated engineering activities and technical training.
- Winner of the Think Award.

ENGINEERING PROJECTS

Experimental Rocket Development

- CAD design of structural and propulsion components.
- Structural simulations using FEA and aerodynamic simulations using CFD.
- Manufacturing planning, prototype assembly, and mechanical integration of propulsion and recovery systems.

TECHNICAL SKILLS

CAD: SolidWorks (4 years of experience), Fusion 360, Onshape

Engineering: Mechanical Design, CAD Modeling, Engineering Drawings, Mechanical Assemblies, Design for Manufacturing (DFM), Prototype Development, FEA, CFD

Manufacturing: CNC Machining Fundamentals, Conventional Machining, Mechanical Assembly, Manufacturing Processes

Programming: MATLAB

Office: Excel, Word, PowerPoint

EDUCATION

Tecnológico de Monterrey — B.S. in Mechanical Engineering — Expected Graduation: July 2029

AWARDS

- FIRST Robotics Competition Mexico National Champion (2025)
- FIRST World Championship Qualifier
- Third Place — NASA International Air and Space Program
- Think Award — FIRST Tech Challenge
- Best Autonomous Robot Award
- Best Control System Award (×2)

LANGUAGES

Spanish — Native English — B2